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Development of a Browser Add-on for the Automatic Translation of Web Texts into German Easy Language

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Abstract

This master's thesis presents the development of a browser add-on designed to automatically translate web content into German Easy Language to improve accessibility for individuals with cognitive impairments. Many users face challenges when navigating complex online content, where Easy Language versions are often missing or of low quality. The add-on utilizes LLMs provided by APIs from Groq Cloud (Llama 3.1 70B) and OpenAI (GPT-4o and GPT-4o Mini) to generate translations, based on a prompt that incorporates all relevant Easy Language rules for online content. Along with its minimal technical demands on the user's device, the add-on requires no user interaction, as it runs automatically in the background without a user interface. A selection of 40 websites, including government sites and news articles, was translated and evaluated using four readability metrics (LIX, Amstad/Flesch, 1. Wiener Sachtextformel, and gSmog). While the translations improved readability, they also resulted in increased text length due to the direct translation approach. Additionally, issues such as inconsistent formatting and unusual symbols were identified. The add-on shows potential as an accessibility tool but requires further development to produce concise translations of high quality.

Keywords: easy language, german easy language, leichte sprache, web accessibility, browser add-on, browser extension, large language models (LLM), readability, automated website translation, API integration

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Chapter 1

Introduction

Nowadays, the internet is an integral part of our daily lives. Whether it is reading the news, booking appointments, handling bureaucratic tasks or general communication: more and more aspects of our lives are shifting to the digital world. Access to the internet has become essential for active participation in today's society.

However, there is an underestimated number of people in Germany who have difficulties accessing online content. Those particularly affected include individuals with mental disabilities (1.8 million people, according to the Federal Statistical Office of Germany [1]), low literacy (6.2 million people, according to the LEO study [2]) or dementia (1.8 million people, according to the German Alzheimer Society [3]). They are faced with barriers that are still widespread on many websites. Confusing structures, hard-to-read fonts and complex texts. Nevertheless, these individuals have the right to access information without barriers. This is established in the Federal Act on Equal Opportunities of Persons with Disabilities (Behindertengleichstellungsgesetz, BGG)[4] and its implementing regulation for accessible information technology (Barrierefreie-Informationstechnik-Verordnung, BITV 2.0)[5].

According to this regulation, all federal agencies are required to provide their websites in Easy Language. However, the reality is quite different: Many agencies do not offer any texts in Easy Language at all, and the quality of the available content is inadequate, which was observed by the study "On The Prevalence of Leichte Sprache on the German Web" [6]. In many cases, the available texts also fail to cover the full range of information. Typically, there is only one page dedicated to Easy Language, and the content on this page is limited. It generally includes instructions on how to navigate the website, which is still written in difficult-to-read language, as well as some basic information about the federal agency or ministry itself. This

is primarily due to the fact that translating texts into Easy Language is both time-consuming and costly, and requires specialized expertise. Often, translators lack the necessary professional training and experience to produce high-quality texts in Easy Language.

The objective of this master thesis is to develop a browser add-on capable of automatically translating text content from any website into German Easy Language. This will involve using a Large Language Model (LLM) in combination with specially designed prompts implementing the rules of Easy Language. The aim is to achieve a translation quality that is comparable to manually translated texts.

A central focus of the technical section of this thesis is to ensure optimal performance of the add-on. The thesis will determine which approach, integrating a locally deployed LLM or leveraging an API connection, offers better possibilities in terms of integration feasibility and accessibility for users. To maximize ease of use, the browser add-on will not include a user interface.

The quality of the translated texts will primarily be evaluated using readability metrics, such as the Amstad formula, and through comparison with manually translated existing pages. This thesis is structured into eight main chapters, each addressing a specific aspect of the research and implementation of the browser add-on. It begins with an introduction that outlines the problem statement and motivation behind the research, with a focus on the importance of Easy Language for accessibility and the potential of LLMs to enhance automated translations. Chapter 2 establishes the theoretical background necessary to understand the research by defining Easy Language, discussing practical guidelines for translation, explaining the translation process, and providing an overview of LLMs. An overview on existing research and technologies relevant to automatic translation into Easy Language is provided in Chapter 3.

The methodology presented in Chapter 4 details preparatory steps before implementation, including prompt definition, LLM integration, model selection, and selection of relevant online content for translation. Chapter 5 then follows with an explanation of the technical aspects and details of the developed browser add-on. This includes describing the system architecture and key components. The evaluation in chapter 6 examines both the quality and readability of the translations. The chapter begins by outlining the evaluation criteria and procedure, offering a detailed explanation of the readability metrics and their application within an automated evaluation script. It then analyzes and compares the results across the three LLMs, as well as between GPT-4o and official Easy Language translations. Additionally, an analysis of the

translations' content quality and a discussion of the most significant observations, such as formatting errors and structure of the content, is included.

The conclusion, see Chapter 7, summarizes the main findings and highlights the potential of LLMs aiding in translating text into Easy Language. Finally, the conclusion is followed by an outlook in Section 7.1, which provides suggestions for future improvements, such as new features regarding the visual presentation of translations. Additional supporting materials are included in the appendices, which list German Easy Language rules (see Appendix A) and details on the evaluated websites (see Appendix B and C).

Chapter 2

Foundations

This chapter covers the foundations of Easy Language and Large Language Models (LLMs). It begins with an introduction to Easy Language, including its history, guidelines, and the process of translation. The chapter then moves on to the foundations of LLMs and provides a brief overview of how to write effective prompts.

2.1 An Introduction to Easy Language

Easy Language is a variation of natural language aimed at making information as understandable as possible [7]. It is characterized by various linguistic and visual features: reduced vocabulary and sentence structure, large font size, one sentence per line, and the visual separation of long words with a hyphen or a point called “mediopoint” (“Mediopunkt”, “.”).

The target audience for texts in Easy Language are primarily individuals who struggle with the original language version and therefore prefer a “simplified” version. This group mainly includes people with cognitive or sensory impairments, who often refer to themselves as “people with learning difficulties” [7]. Other target groups include individuals with dementia, prelingually deaf individuals, and those affected by functional illiteracy. According to Maaß [7], migrants learning German as a second language can also benefit from texts in Easy Language. As this group often needs to handle administrative matters in the early stages of migration, Easy Language can provide significant relief.

The concept of Easy Language originally comes from Finland, where it was first introduced in the 1970s. A few years later, in 1988, the organization Inclusion Europe was founded, which has advanced the Easy Language movement across Europe [8]. This includes the creation of guidelines and a checklist for self-assessing the clarity of texts [9]. In Germany, the BGG established the first legal framework for accessibility in 2002. In addition to the literal meaning of accessibility, which refers to access to buildings and transportation, it also addresses communication barriers. Easy Language is intended to serve as a tool for ensuring access to information and communication facilities.

With the introduction of the “Ordinance for the Creation of Barrier-Free Information Technology under the Disability Equality Act”, known as the “Accessible Information Technology Regulation” (*Barrierefreie-Informationstechnik-Verordnung*, BITV 2.0), in 2011, the requirements of the BGG were specified. The ordinance mandates that federal websites must be available in Easy Language.

2.2 Practical Guidelines of Easy Language

There are several guidelines for translating into Easy Language, which have either evolved from practice or been established through legal requirements over time. The three most well-known guidelines in Germany are:

- The guidelines from Inclusion Europe, originally written in English and translated into many other languages [10]
- The guidelines from the Netzwerk Leichte Sprache [11]
- The 13 rules of BITV 2.0, which pertain to texts on the internet [5]

Overall, there are 120 rules, 17 of which are consistent across all three of the mentioned guidelines [12]. Table 2.1 offers a concise summary of these shared rules, which address different linguistic and design aspects, including word level, sentence level, text level, as well as character level, typography, and layout.

However, since the rules of BITV 2.0 are particularly incomplete and unsystematically presented, the Research Center for Easy Language at the University of Hildesheim has committed itself to review all guidelines based on linguistic principles [7] and provide a revised and precise version of the rules. This process involves collaboration with a reference group of prelingually deaf individuals.

Category	Rule
Typography and Layout	1. Larger font size 2. Each sentence on a new line 3. No word breaks at the end of lines 4. Text is left-aligned
Word Structure	5. Short words 6. Separation of compound words with hyphens 7. No abbreviations 8. No passive constructions
Vocabulary	9. Simple words 10. Preferably no foreign words 11. Foreign words are explained where they are needed
Sentence Structure	12. Short sentences
Semantics	13. No negations
Text	14. No lexical variation in the text: same designation for the same concept 15. Relevant information first 16. Clear structure: subheadings are used 17. Readers are addressed directly

Table 2.1: Common Rules from All Three Guidelines (Based on [13])

Therefore, this master's thesis will rely on the rules developed by the Research Center to ensure high-quality implementation.

The rules summarized in Table 2.1 provide a comprehensive overview of the concept of Easy Language. However, they are criticized by the Research Center for Easy Language:

Rule 12 specifies that sentences must be short. Yet it is unclear how this should be implemented and what exactly qualifies as a short sentence. The real issue lies more in the complexity of sentence structure than in the actual sentence length. Even short sentences can be difficult to understand due to complex nominal structures [12]. These ambiguities are present in many of the rules and pose more of a hindrance than a support for translators.

To keep the scope of this chapter manageable, a detailed description of each rule in the guidelines and their revised definitions by the Research Center has been excluded. The full set of revised rules for German Easy Language can be found in Appendix A.

The following example is an Easy Language article published by the German public radio and television broadcaster NDR ¹, which reports on the strikes at Deutsche Bahn. It clarifies the meaning of a strike, and outlines the demands of the train drivers. With its structure, simple vocabulary and explanation of difficult words it effectively demonstrates the characteristics of an Easy Language text:

Bei der Bahn waren in den vergangenen Wochen viele Streiks.

Bei einem Streik arbeiten Menschen **nicht**.

Diese Menschen sind zum Beispiel mit ihrer Arbeit **nicht** zufrieden.

Sie wollen zum Beispiel mehr Geld verdienen.

Oder sie wollen weniger Stunden arbeiten.

Diese Streiks waren von Lok-führern.

Deshalb sind viele Züge ausgefallen.

Und deshalb waren viele Probleme im Zug-verkehr.

Aber jetzt haben sich die Lok-führer und die Bahn geeinigt.

Die Lok-führer haben gesagt:

Wir bekommen mehr Geld für unsere Arbeit.

Und wir können weniger Stunden arbeiten.

Deshalb sind wir sehr zufrieden.

Und deshalb streiken wir in den nächsten 2 Jahren **nicht** mehr.

...

2.3 The Process of Translation into Easy Language

Translating a text into Easy Language is often more challenging than creating a text directly in Easy Language. When writing an original Easy Language text, the author controls both the content and its presentation. In contrast, translating a source text involves handling a higher density of information.

This presents the translator with two possible strategies:

1. Retain all the information from the source text and explain all implications to meet the needs of the target audience, or

¹https://www.ndr.de/fernsehen/barrierefreie_angebote/leichte_sprache/Einigung-bei-der-Bahn,lokfuehrer226.html

2. Include only the most relevant information to balance text length and comprehensibility.

These strategies create a dilemma between readability and the right to information, posing a challenge for the translator [7]. Usually, the translator makes a reasonable compromise between these two approaches in coordination with the client. After consulting the client responsible for the source text, the main statement is determined. Any information that is peripheral will not be taken into account when translating into Easy Language. Depending on the topic of the source text, a longer target text can also be accepted to meet the information demand of the target audience. All in all, the translation process requires a high degree of communicational competence and clearance before starting with the actual translation. As consolidated by Maaß [7], the following steps form the translation process:

1. **Analyze the source text: How is the text structured, and what challenges does it present?**
2. Consult with the client.
3. **Identify the target audience.**
4. Develop a translation strategy with the client.
5. **Translate the source text.**
6. Revise the target text.
7. **Format the document.**
8. Record the developed explanations and translation solutions in a database.

All steps marked in bold (1, 3, 5 and 7) can be handled by the add-on's automatic translation. The remaining steps, which involve client interaction and translation review, will need to be skipped.

The first step of the process includes analyzing the various dimensions of the source text. There are situational, intertextual, functional, thematic, and linguistic dimensions that have to be considered before translating. This is an important step not explicitly covered by the practical guidelines, but it still needs to be considered when developing the prompt for the LLM.

Another step, which is not listed in the above process, is the review of the translation by the target groups. According to the rules of Netzwerk Leichte Sprache [11], a

text can only be considered an Easy Language text if a target group review has been conducted. This requirement is understandable, but also detrimental to the important goal of making as much content as possible accessible in Easy Language. The Forschungsstelle Leichte Sprache Hildesheim [12] criticizes that a review by target groups is an excellent way to demonstrate inclusion and can be a criterion for a quality label (e.g., by Inclusion Europe). However, it should not be a requirement for classifying a text as Easy Language. Just as a translation into any other national language does not need to be approved by a group of people of the target nationality to be considered a valid translation.

This perspective helps diminish doubts about automatic translation, emphasizing that while target group reviews are ideal, they aren't always feasible – even when translating manually.

2.4 Brief Introduction to Large Language Models (LLMs)

The following section provides a brief overview on Large Language Models derived from Google for Developers [14].

As part of the broader family of language models, LLMs are designed to understand and generate human language based on patterns in text data. While early language models were limited to predicting the probability of a single word, modern LLMs are capable of generating entire paragraphs or even complete documents.

LLMs earned their name because of their massive scale, characterized by billions of parameters and extensive training on large datasets, which together enable them to handle complex natural language processing (NLP) tasks with high accuracy. Those tasks go beyond the LLMs original task, which is generating the most plausible text in response to an input. With their so called emergent abilities, LLMs are now beginning to show strong performance on tasks like translation, question answering, or even writing code. As a subset of LLMs, Transformers are the state-of-the-art architecture for such tasks. A major drawback of using models this size is the increased computational cost and storage requirements, which raises the central question for implementing the browser add-on: How can an LLM be integrated in a performance-efficient way?

Another aspect to consider when applying LLMs for NLP tasks is prompt engineering, which is also covered by Google for Developers [15]. The aim is to ask the right question to get the best possible output from an LLM. No advanced programming knowledge is needed to interact with such a model; simple, plain language prompts are sufficient. Various best practices have been established already, such as structuring the prompt, giving examples and using constraints. An optimal prompt structure includes defining the role of the LLM, giving context and then providing the instruction. There are different types of prompts, such as direct prompting (Zero-shot), which is the simplest form and involves providing only the instruction or setting a role without any examples. Figure 2.1 illustrates this type of prompt, showing how a task is presented with just the instruction and no additional context.

Prompt:

Create a four-column spreadsheet of 10 highly-rated science fiction movies, year of release, average audience rating, and top 3 keywords from audience reviews.

Make sure to cite the source of the audience rating.

Figure 2.1: Zero-Shot Prompt Example [15]

In contrast, one- or few-shot prompting includes one or more specific examples to demonstrate what is expected from the model's output. A few-shot prompt can be structured as in Figure 2.2, where examples of input and expected output are provided after defining the task.

Generate a rhyming couplet about a tree:

Input: "cat"

Output: The curious cat, so sleek and fat, Curled up cozy on the welcome mat.

Input: "sky"

Output: Look up high into the endless sky, Where birds and clouds go drifting by.

Input: "tree"

Output:

Figure 2.2: Few-Shot Prompt Example [16]

Applying this knowledge of prompting is crucial for achieving optimal results in translating web texts into Easy Language.

Chapter 3

Related Work

In the field of web accessibility and browser add-ons, solutions like Easy Reading [17], WebHelpDyslexia [18], and others have emerged to assist users with specific needs. Providing the user with tools to adapt the font, color or formatting of web content. Along with other accessibility features, these add-ons ensure that online information is more inclusive. Compared to the manual content adaptation, this thesis aims to develop a browser add-on with automated translation of web content into Easy Language. While the main focus is the use of an LLM for the translation task, there will also be other automated aspects like the change of fonts and layout according to Easy Language guidelines.

With the rise of Artificial Intelligence (AI) and the popularity of chatbots, its integration into various fields has become increasingly prevalent. This includes exploring how AI can be utilized to improve accessibility, as demonstrated by a recent feasibility study on using ChatGPT¹ for translating texts into Easy Language by Deilen et al.[19]. The study was conducted in April 2023 and the results are based on GPT-3.5. They used twenty texts from different websites of German public agencies, with each text containing between 179 and 672 words about citizen-oriented topics.

These texts were simplified by testing two approaches. The first approach was to repeatedly ask the chatbot to translate the text into German Easy Language and to simplify the text, without giving further instruction. The second approach included multiple steps to simplify the texts, based on the strategies that are applied in Easy Language translation. Each prompt represented a simplification on text, sentence and word level.

The final outputs of both approaches were then compared on correctness and

¹<https://chatgpt.com>

readability. The correctness was evaluated independently by two people and the readability was assessed using the TextLab software², which determines the text-comprehensibility based on the Hohenheim Comprehensibility Index (HIX). The HIX takes four major readability formulas into account, which is also discussed in Chapter 6. The syntactic complexity was determined using the Stanford NLP Python Library Stanza, with the assumption that the higher the number of clause dependents, the more complex the texts. Altogether, 37.5% of the generated texts contained correct content, with the second approach producing better results (55% correct) than the first (20% correct). To be classified as a German Easy Language text, the benchmark is set at an HIX value of 18. The first approach yielded a mean HIX value of 15.3 and the second a value of 9.5 points. Still, some texts of the first approach reached an HIX value of at least 18 points.

All in all, the results show that the generated texts are easier to read than the source texts, but still do not meet the Easy Language standards. The authors conclude that ChatGPT can be used as a template for professional translators, rather than as a standalone translation tool. They also point out that more precise prompts, such as assigning a role to ChatGPT and setting the context, can improve the comprehensibility of the generated texts.

As this thesis focuses on the use of LLMs for automatic translation within a browser add-on, initiatives like WebLLM [20] become particularly relevant. WebLLM is an efficient in-browser LLM inference engine that operates entirely within the browser, leveraging WebGPU for hardware acceleration, without relying on server-side processing. This approach not only reduces cost, but also enhances privacy and personalization. It also offers features like real-time interactions and streaming outputs, and can be integrated into a variety of applications, including browser add-ons.

²<https://www.comlab-ulm.de/textlab-sprachsoftware/>

Chapter 4

Methodology

This chapter outlines the methodology behind developing the browser add-on. It covers essential aspects such as prompt design, the integration of the LLM into the add-on, and the selection process for the model, all of which are crucial for the technical implementation. Additionally, the chapter addresses the process of selecting relevant online content for evaluating the add-on's output.

4.1 Defining the Prompt

As described in Chapter 2.4, this thesis uses a few-shot prompting strategy, which is well suited for the project's goals. The prompt starts by defining the LLM's role and the task's context. Here, the LLM is set up as a translator for German Easy Language, with a focus on ensuring compatibility with HTML formatting, as the translations are intended for websites. Following this setup, the prompt includes a clear task description and a list of Easy Language rules specifically relevant to online content. A shortened version of this prompt is shown in Figure 4.1.

The rules for Easy Language translation are derived from the work of the Research Center for Easy Language at the University of Hildesheim (see Appendix A), except for those that are not applicable to online content or the nature of the AI task, such as the use of pictures for illustration. Since the translation is automatically generated and not reviewed, incorporating images could lead to issues with ownership rights. Additionally, there is a risk that the pictures could be confusing or not accurately represent the intended meaning. Therefore, the rule regarding the use of illustrations is excluded from the prompt to avoid these potential complications. To structure the prompt as a few-shot prompt, each rule is accompanied by a positive and negative

example. The input text is added after the final instruction to carefully review the translation and formatting before delivering the final result. This part was included after reviewing multiple translations that contained common formatting errors. All in all, this prompt is the outcome of several revisions, testing various orders of text and instructions.

###Rolle###

Du bist ein Übersetzer, der Texte von regulärem Deutsch in Leichte Sprache umwandelt. Dein Ziel ist es, den Text so zu vereinfachen, dass er für Menschen mit kognitiven oder sprachlichen Einschränkungen leicht verständlich ist.

###Aufgabe###

Übersetze den folgenden Text in Leichte Sprache. Orientiere dich dabei an diesem Regelwerk, welches sowohl Negativ- als auch Positivbeispiele aufzeigt. Wende die Regeln Schritt für Schritt an:

[Easy Language Rules]

###Denk daran###

Prüfe dein Ergebnis sorgfältig auf Verständlichkeit gemäß der Regeln. Der Leichte Sprache Text wird den vorherigen Text ersetzen. Ergänze den Mediopunkt bei langen Worten. Liefere nur den übersetzten Text, nichts anderes. Füge am Ende jeder Zeile und nach einem Doppelpunkt das HTML-Tag `<br>` ein. Und nutze Fettdruck mit dem HTML-Tag `<strong>` und keine `**` gemäß den Regeln.

###Übersetze den folgenden Text in Leichte Sprache###

[Text]

Figure 4.1: Prompt for the Translation into German Easy Language

Another important step, handled separately from the prompt, is the inclusion of a disclaimer at the end of each translated text on the website. This disclaimer informs users that the translation was automatically generated by a computer and has not been human-reviewed. By including this warning, users are reminded of the potential variability in translation quality and correctness of the content.

4.2 Integration of the LLM

During the development of the browser add-on, two strategies for integrating the LLM were evaluated. The first approach involved local integration through

the WebLLM library [20], but this proved to be challenging due to the technical requirements, which not all user devices could meet. Although local integration offers advantages such as reduced latency from processing directly on the user's device, these benefits are outweighed by technical challenges. These include the need for advanced hardware (such as WebGPU) and the additional storage requirements, as the LLM must be initially downloaded beforehand. Testing local integration with the WebLLM Library's example code [21] showed that each device encountered different problems. Some devices failed to download the LLM due to timeouts, while others lacked the necessary WebGPU support. These inconsistencies made local integration unreliable. The same issues can be replicated when attempting to use the chat function provided by the WebLLM developers¹.

Furthermore, updating and maintaining the library on various devices could be a complex process. An argument for local integration could be improved privacy, but this is not a concern in this case, as the add-on only uses publicly available content for its prompt, with no user-specific information required or processed by the LLM.

The second strategy, using an API connection to access the LLM, proved to be more viable. The API connection simplifies many of the technical challenges associated with local integration. By relying on a cloud-based service, it ensures that the latest version of the model is always available to users without requiring device-specific processing power or storage. The ease of deployment and scalability makes this approach suitable for wide accessibility. However, it requires an API key, which presents a challenge in terms of cost and management. For the published version of the add-on, the most practical solution would be to provide users with an API key via a server (at a cost), though this solution adds a layer of complexity in terms of key distribution and management.

Currently, for testing purposes, the browser add-on allows users to input an API key for GroqCloud API², which supports several popular LLMs such as Llama 3.1 by Meta³ and Gemma by Google⁴. Additionally, the OpenAI API⁵ for GPT models is available, although it comes with usage limits and a cost. The GroqCloud API is provided for free, but it has a limitation on the number of calls that can be made per minute. The OpenAI API, while available for use, imposes both usage limits and associated costs. This constraint may impact the performance and responsiveness of

¹<https://chat.webllm.ai>

²<https://console.groq.com/docs/overview>

³https://github.com/meta-llama/llama-models/blob/main/models/llama3_3/MODEL_CARD.md

⁴<https://huggingface.co/google/gemma-2-9b-it>

⁵<https://platform.openai.com/docs/overview>

the add-on, particularly with high traffic or frequent usage.

4.3 Model Selection

To translate the online content of websites, three LLMs were selected for evaluation:

- Llama 3.3 70B ⁶
- GPT-4o ⁷
- GPT-4o mini ⁸

These models were chosen based on a combination of critical factors. Llama 3.3 70B is an open-source model accessible via the Groq API, offering a cost-effective alternative to proprietary models. Optimized for multilingual dialogue use cases, it is particularly well-suited for German Easy Language translation compared to other open-source models.

With ChatGPT being the most widely recognized LLM among the general public and in research, the current flagship model GPT-4o (“o” for “omni”) was also selected for evaluation. The inclusion of GPT-4o mini offers an opportunity to assess the balance between computational efficiency and performance when using a more compact version of the same model.

Given the high API costs of GPT-4o, comparing it to Llama 3.3 70B is particularly relevant when considering the cost implications for potential publication of the add-on. As an open-source model, Llama can be hosted on a private server, reducing expenses and offering opportunity for fine-tuning.

4.4 Relevant Online Content for Translation

A variety of online content was chosen for the translation task, consisting of overall 40 websites. The evaluation based on these websites is divided in two phases. In the first phase, 20 websites are translated by three selected LLMs. These include 10 German federal websites, which focus on sociopolitical issues and provide information

⁶https://github.com/meta-llama/llama-models/blob/main/models/llama3_3/MODEL_CARD.md

⁷<https://platform.openai.com/docs/models#gpt-4o>

⁸<https://platform.openai.com/docs/models#gpt-4o-mini>

about the German political system, as well as 10 news websites from Tagesschau and Hessenschau, covering topics like weather, politics, social issues, and health. The corresponding links to these websites are listed in Appendix B.

The selection of these websites was based on their significance and accessibility standards. In Germany, federal websites must provide content in Easy Language to ensure it is comprehensible to all citizens, as mandated by the BITV 2.0 [5]. In addition, major news outlets like Tagesschau ⁹ are embracing their responsibility to provide accessible content by offering a live television segment presented in simple language.

In the second phase, the LLM that performed best in the first phase is used to translate another set of 20 websites, which includes both federal and additional news sites. The translations are then compared to the Easy Language versions provided by the websites themselves. This approach narrows the selection down to content from news platforms such as NDR ¹⁰ and MDR ¹¹, as well as press releases from federal websites. All selected websites vary in content length and complexity to ensure a diverse basis for evaluation.

⁹<https://www.tagesschau.de>

¹⁰<https://www.ndr.de/index.html>

¹¹<https://www.mdr.de/nachrichten/index.html>

Chapter 5

Implementation

This chapter will discuss the technical architecture of the browser add-on, along with implementation details and source code references. The browser add-on is developed specifically for Chrome due to technical reasons related to manifest versioning. Manifest version 3, which is the latest standard, offers improved security and performance but is not yet fully supported by Firefox. At the same time, Chrome is deprecating the previous version (manifest version 2). Since Firefox does not fully support version 3, and to ensure compatibility with the latest web standards, the decision was made to focus on a version for Chrome only, rather than a hybrid add-on for both browsers.

5.1 Technical Architecture

The browser add-on closely follows the standard implementation practice for Chrome Extension development¹. In the `manifest.json` file, key meta-information is defined, such as the extension's manifest version, name, and icon. As mentioned earlier, this project is built using manifest version 3, the latest specification for Chrome extensions. The `manifest.json` file also specifies the scripts responsible for managing the popup, content manipulation, and background tasks. To control where content scripts are injected, a list of URLs can be defined for exclusion. In this case, all search engine-related content is excluded from translation by the add-on, as translating this type of content negatively impacts performance and is not relevant for the user.

¹<https://developer.chrome.com/docs/extensions/develop>

As illustrated in Figure 5.1, the `content.js` script plays a key role in interacting with the webpage by accessing its DOM and the browser's local storage. Using message passing², the `content.js` script exchanges information with the service worker `background.js`. A service worker is a background script that operates independently of the webpage and handles tasks such as API communication and managing background functionality. The `background.js` script receives messages containing webpage content from `content.js` and interacts with external APIs, enabling access to the selected LLM for the translation tasks.

The API client and model to be used are configured via the browser's popup interface, defined by `popup.html` and `popup.js`. This popup, shown in Figure 5.2, provides fields for settings such as the API key, a checkbox to activate debug mode, and the translation function. Figure 5.2 also displays the icon of the add-on, which depicts a person reading a book. This icon is similar to those commonly used on websites to switch to Easy Language content. Dropdown menus allow users to select the client (e.g., Groq Cloud or OpenAI) and dynamically update the model options based on the chosen client. It is important to note that these settings are primarily intended for developers and are designed for testing purposes. For example, entering an API key manually is a temporary requirement that will be removed in the future. In the final version of the add-on, users will not need to input an API key themselves, as this process should be handled automatically by the browser add-on.

All user-defined settings are stored in the browser's local storage, ensuring that the `content.js` script can access them and pass the necessary information to the service worker. Additionally, the local storage is utilized to save the translations of the current webpage, preventing redundant API calls upon page reload. The translations are stored as a variable, which is structured like a dictionary, where the original website text serves as the key, and the corresponding translation is the value.

²<https://developer.chrome.com/docs/extensions/develop/concepts/messaging>

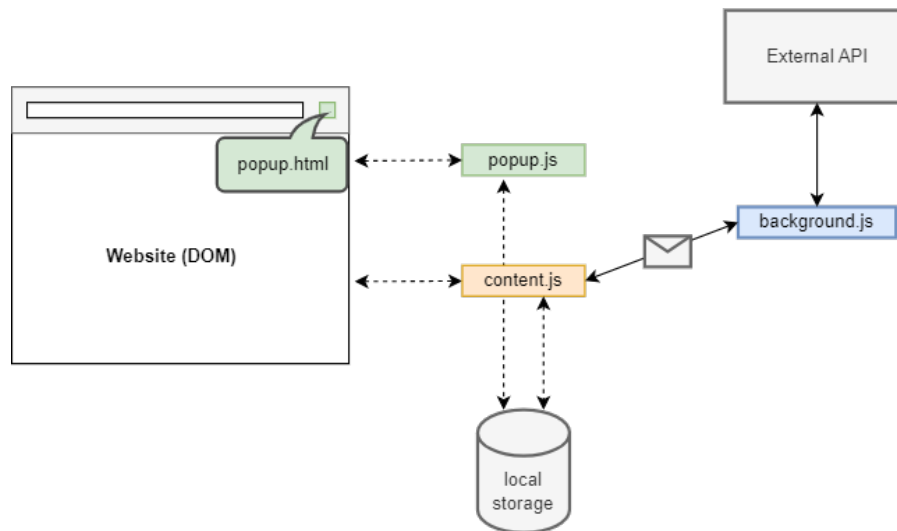


Figure 5.1: Architecture of the Browser Add-On

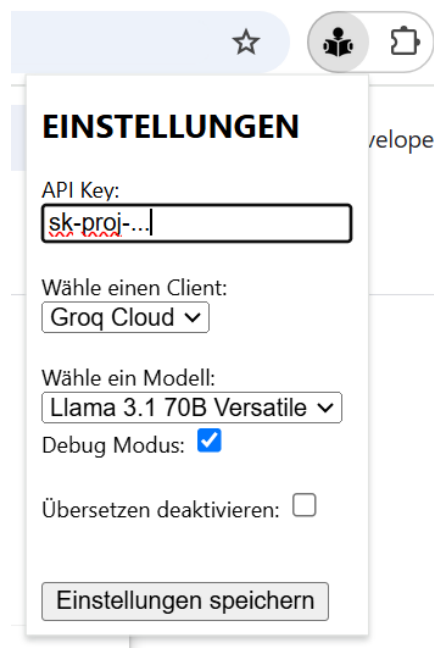


Figure 5.2: Popup for Settings of the Browser Add-On

5.2 Implementation Details

The main process of the implemented browser add-on focuses on translating web-site content into Easy Language using asynchronous functions and caching to ensure optimal performance. As illustrated in Figure 5.3, the browser triggers the

DOMContentLoaded event when a webpage finishes loading. This event signals the content script to execute its primary function, `translateMainContentAsync` (more details on its implementation in Section 5.2.1). This function first checks the browser's local storage to determine whether translations for the relevant content already exist. If translations are found, they are retrieved from the cache and applied directly to the webpage content. If translations are not available in local storage, the content script passes a message to the background script, which forwards the request to the external API, see Section 5.2.2 for details. The background script sends the API's translation to the content script, where it is stored in the browser's local storage for future use, and the webpage content is updated in real time. If debug mode is enabled in the settings, the translations are also downloaded as a JSON file. This file contains key information for evaluation, such as the LLM used for translation and the URL of the translated webpage.

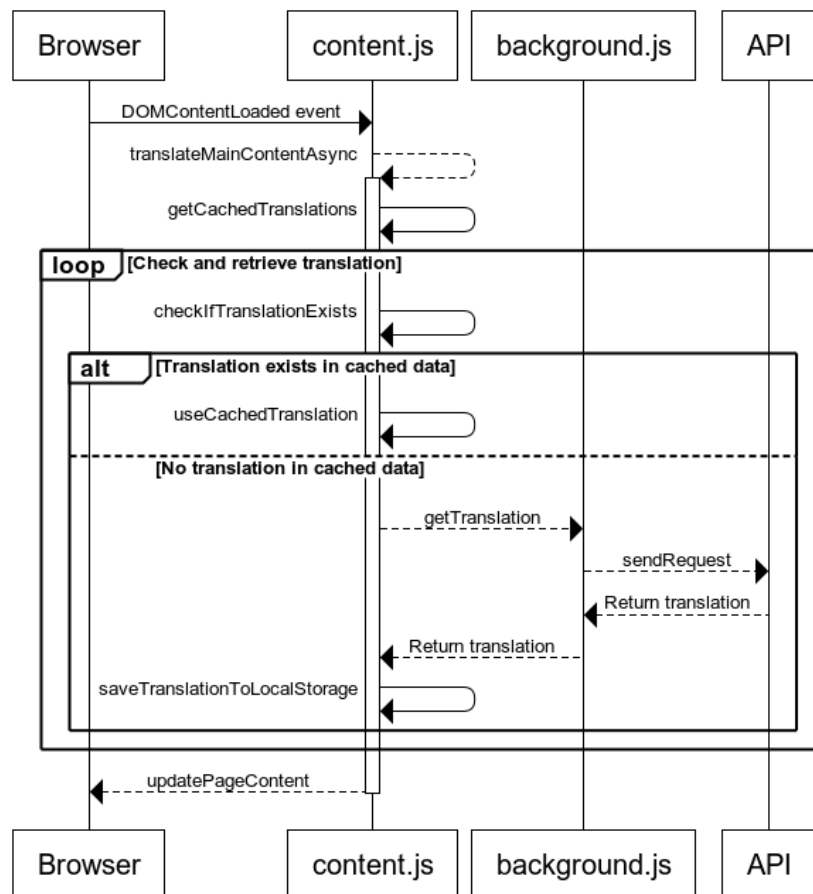


Figure 5.3: Simplified Workflow of the Implemented Translation Process

5.2.1 Content Script (content.js)

As part of the previously mentioned `translateMainContentAsync` function, the `isContentRelevant` function, shown in Listing 5.1, plays a crucial role in the browser add-on's main logic by filtering out irrelevant DOM elements for translation. It checks various criteria, such as whether the element belongs to the header or footer, if the text is too short, or if it involves Table of Contents (TOC) or citation-related content. Only the `<p>`, `<div>`, and `<li>` elements of the webpage are considered for translation. Headings remain unchanged to avoid inconsistencies with references that might appear elsewhere on the website. By excluding unnecessary content based on these conditions, the function not only improves the efficiency of the translation process, but also prevents the waste of tokens, which are words and characters processed by the LLM. This is particularly important because LLM usage is constrained by a token per minute (TPM) limit, which is further managed by the logic of the service worker.

```

1 const isContentRelevant = (content, element) => {
2   const wordCount = content.split(/\s+/).length;
3   const isInHeaderOrFooter = element.closest('header, footer, [class*="
      footer"], [id*="footer"], [class*="header"], [id*="header"], [class
      *="teaser"], [class*="menu"], [class*="nav"]') !== null;
4   const isShortText = content.length < 50;
5   const isShortWordCount = wordCount < 5;
6   const hasTOCClass = element.className.includes('toc') || element.id.
      includes('toc');
7   const isReferenceOrCiteNote = element.id.includes('cite');
8
9   return !isInHeaderOrFooter && !isShortText && !isShortWordCount && !
      hasTOCClass && !isReferenceOrCiteNote;
10 };

```

Code Listing 5.1: JavaScript Function for Content Filtering

Since the LLM is provided with the rendered text without HTML during translation, all hyperlinks in the original texts are lost. The function `restoreLinksInText`, shown in Listing 5.2, resolves this issue by inserting the links back into the translated text. It iterates over all `<a>` tags in the original DOM element, extracting the visible link text (`linkText`) and the URL (`linkHref`). It then replaces occurrences of `linkText` in the translated content with properly formatted `<a>` tags that contain the original URLs.

```
1 const restoreLinksinText = (translation, element) => {  
2   const links = element.querySelectorAll('a');  
3  
4   links.forEach(link => {  
5     const linkText = link.innerText;  
6     const linkHref = link.href;  
7     translation = translation.replace(linkText, '<a href="linkHref">{  
8       linkText}</a>');  
9   });  
10  return translation;  
}
```

Code Listing 5.2: JavaScript Function Restoring Link References in the Translated Text (content.js)

5.2.2 Service Worker (background.js)

The service worker's main functionality invoked by the content script is implemented in the `fetchTranslation` function, shown in Listing 5.3. Before sending a request to the API in line 29, the function first checks whether the token limit is reached. The limit is different for each available LLM and saved in the `constants.js` file. The function estimates the token count by splitting the prompt into words and updates the global variable `remainingTokens` once the translation is fetched. The lock flag ensures atomicity in these asynchronous calls by preventing multiple requests from simultaneously modifying the `remainingTokens` variable. While the lock is active, other requests are paused to avoid race conditions.

```
1 const fetchTranslation = async (text, modelName, clientName, apiKey) => {  
2   const prompt = definePrompt(text);  
3   // Estimate tokens required for the prompt  
4   const tokensRequired = prompt.split(/\s+/).length;  
5   const tokenLimitPerMinute = getTokenLimit(modelName, clientName);  
6   while (true) {  
7     // Wait for lock on token count to be released  
8     while (lock) {  
9       // Wait for 30ms before retrying  
10      await new Promise(resolve => setTimeout(resolve, 30));  
11    }  
12    lock = true;  
13  
14    if (canSendRequest(tokensRequired, tokenLimitPerMinute)) {  
15      // Update number of used tokens
```

```
16     remainingTokens -= tokensRequired;
17     // Release lock
18     lock = false;
19     break;
20 }
21 // Release lock
22 lock = false;
23 console.log("Token limit reached. Waiting for reset...");
24 // Wait for 60 seconds before checking again
25 await new Promise(resolve => setTimeout(resolve, 60000));
26 }
27 try {
28     let client = getClient(clientName, apiKey);
29     const result = await sendRequest(client, prompt, modelName);
30     return result;
31 } catch (error) {
32     // Release lock
33     lock = false;
34     throw new Error("Error during API request: " + error);
35 }
36 };
```

Code Listing 5.3: JavaScript Function for Fetching Translations via API (background.js)

Chapter 6

Evaluation

This chapter examines both the readability and the overall quality of the translations provided by the LLMs. Readability is assessed using metrics such as the Amstad formula, average sentence length, and average number of long words to measure how easy the text is to read. These metrics, which help determine whether the simplified language meets accessibility standards, are described in detail in Section 6.1.

To make the evaluation process more efficient, a Python script was developed to automate the calculation of readability scores, as explained in Section 6.2. Additionally, the LLM-generated translations are examined for grammar, formatting, and coherence through spot checks (Section 6.3.3). However, usability testing and user experience assessments are excluded, as the add-on operates entirely in the background without providing a user interface. An overview of the evaluation process is summarized in Figure 6.1, which outlines the steps taken. First, a selection of 20 websites is translated into Easy Language using each of the three LLMs. Comparing the results with the original (Section 6.3.1), the LLM with the best score is then selected to translate an additional 20 websites, this time from a dataset that includes corresponding Easy Language versions of the original texts. These translations are analyzed to assess whether the browser add-on achieves a similar level of readability as the officially provided Easy Language content. The readability scores of these translations are evaluated and compared in Section 6.3.2. Finally, selected translations are spot checked to assess grammar and content quality.

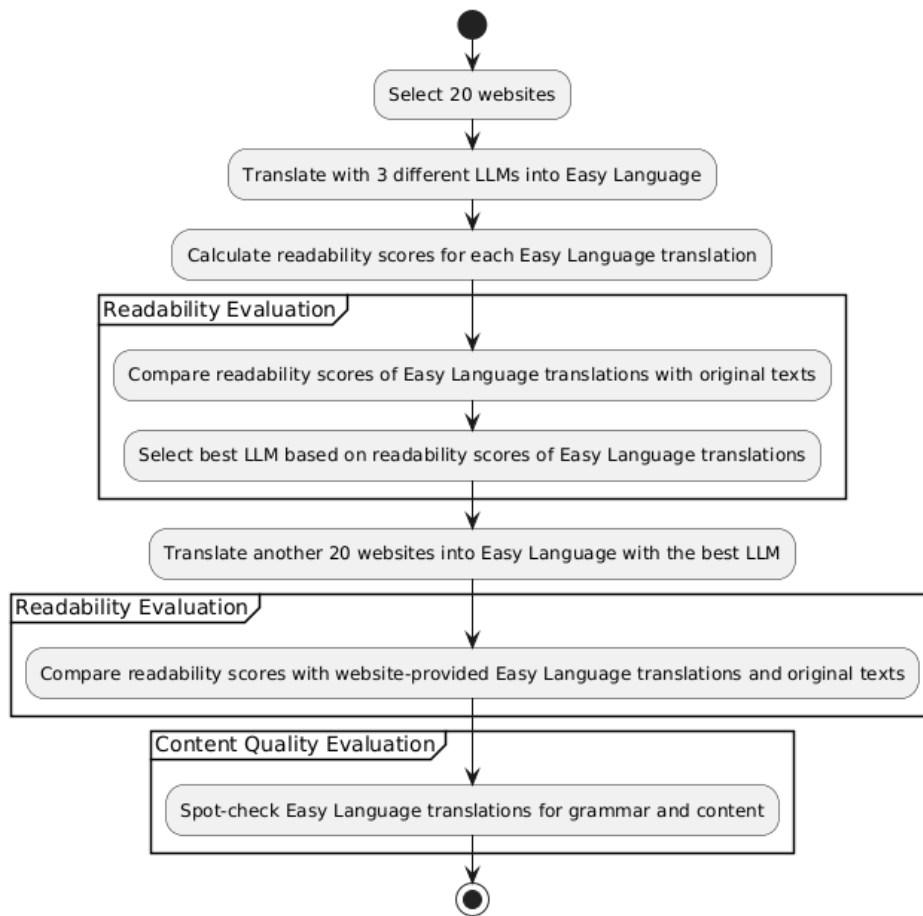


Figure 6.1: Evaluation Process including different readability evaluations, comparing the performance of 3 LLMs and the comparing the results to existing Easy Language texts

6.1 Evaluation Criteria and Procedure

The readability of the Easy Language translations is evaluated using four formulas, which also contribute to the calculation of the HIX [22]. As discussed in Chapter 3, the HIX is the primary metric used to classify Easy Language texts. The precise calculation of this index is implemented in proprietary software called TextLab, which incorporates the four formulas alongside additional statistics, such as average sentence length and the number of words longer than six letters. While it is not possible to directly calculate the HIX for comparison with the result of the study [19] in Chapter 3, these four formulas are the closest available approximation.

The formulas are applied to both the original and translated texts to enable a compar-

ison of their readability. While most formulas are designed for universal use across languages, some are specifically adapted for German to ensure accurate results. They calculate readability scores using statistical analysis, focusing on measurable factors such as word length, sentence count, and the use of complex words. These scores aim to capture “all features of a text that make it easier for a specific group of readers to understand its meaning” [23]. Readability tends to decrease with an increase in the number of difficult words, lexical variation, average sentence length, or subordinate clauses.

Still, some metrics have been adapted to fit the formulas, which is discussed in the following subsections. It should also be noted that these formulas do not assess the actual comprehensibility of the text’s content. As a result, texts with complex content can still achieve a readability score similar to that of simpler texts. To complement the quantitative analysis, some texts are manually spot-checked for content accuracy.

6.1.1 Amstad Formula/Flesch Reading Ease

The Amstad formula derives from the Flesch Reading Ease formula, originally developed for English texts. It incorporates key variables such as word length and sentence length. To account for the generally longer word lengths in the German language, Toni Amstad adapted the original formula to ensure its applicability to German texts [24].

The Amstad score is calculated as follows:

$$\text{Amstad} = 180 - \text{ASL} - (58.8 \times \text{ASW})$$

Where:

- **ASL (Average Sentence Length):** The average number of words per sentence.
- **ASW (Average Syllables per Word):** The average number of syllables per word.

A higher score indicates an easier-to-read text, with a score ranging from 0 to 100. An easy to read text reaches a score starting at 70, while anything below 60 is considered demanding or hard to read (e.g. scientific papers) [25].

6.1.2 1st New Vienna Text Readability Formula (1. WSTF) / 1. neue Wiener Sachtext-Formel

The 1st New Vienna Text Readability Formula is specifically designed by literary researcher Richard Bamberger and Erich Vanecek [26] to evaluate the complexity of German-language texts. There are four different variations of the formula, each assigning different weights to its input variables, based on regression equations. The 1st New Vienna Text Readability Formula is calculated as follows:

$$\text{WSTF}_1 = 0.1935 \times \text{MS} + 0.1672 \times \text{SL} + 0.1297 \times \text{IW} - 0.0327 \times \text{ES} - 0.875$$

Where:

- **MS:** The percentage of words with three or more syllables.
- **SL:** The average sentence length/number of words per sentence.
- **IW:** The percentage of words with more than six letters.
- **ES:** The percentage of monosyllabic words.

It assigns a grade level indicating the educational background required to comprehend the text, ranging from level 4 (very easy) to level 15 (very difficult).

6.1.3 Simple Measure of Gobbledygook – German (gSmog)

Based on McLaughlin's original formula, Bamberger [26] made a few adaptations to better suit the German-speaking context. This adaptation measures the proportion of polysyllabic words (defined as words with three or more syllables) relative to the total number of sentences in a text. As the original formula was designed for analysis based on a sample of 30 sentences, adjustments need to be made to account for differences in text length:

$$\text{gSmog} = \sqrt{\left(\frac{\text{Total Polysyllables} \times 30}{\text{Total Sentences}} \right)} - 2.0$$

The result approximates the reading age (in school grade levels) required to understand the text [26].

6.1.4 Readability Index (LIX)

The LIX (abbreviation of Swedish Läsbarhetsindex) is calculated on the basis of the average sentence length and the percentage of long words in a text. Originally developed by the Swedish researcher Björnsson [27], the score ranges from 15 to 80. An interpretation of the score by Bamberger is documented in Table 6.1, connecting the different ranges to their corresponding school grade level. This interpretation only covers ranges from 15 to 38, with the oldest grade level being 8 (middle school). Since metrics such as LIX are often applied to evaluate the readability score of text books for students, the results are commonly expressed in terms of school grade levels. A more general interpretation is also included in Table 6.1, covering the full range of the score with labels from “very easy” to “very difficult”. The index is calculated based on the average sentence length and the percentage of long words, which are defined as words with more than six letters.

$$\text{LIX} = \frac{\text{Total Words}}{\text{Total Sentences}} + \frac{\text{Total Long Words}}{\text{Total Words}} \times 100$$

	LIX	Grade (according to Bamberger)
Very Easy	15+	1-4
easy	30+	5-8
medium	40+	
difficult	50+	
very difficult	60+	

Table 6.1: Interpretation of the LIX Score (based on [23] and [28])

6.2 Automated Evaluation Script

To automate the evaluation process, a Python script was developed, utilizing the Python Library Readability (v0.3.1)¹, which provides various readability metrics and other statistics. This package was chosen because it supports German texts, offering tools for identifying basic German words, counting sentences, and recognizing polysyllabic words. The readability tool requires tokenized text as input, which is handled by the Stanford NLP Python Library Stanza (v1.9.2)². Stanza offers NLP-based tokenization, part of speech (POS) tagging and dependency parsing for many languages, including German.

¹<https://github.com/andreascv/readability>

²<https://github.com/stanfordnlp/stanza>

The evaluation pipeline is detailed in Code Listing 6.1 and operates as follows: All JSON files containing translations and their corresponding original texts are read, tokenized, evaluated, and the results are saved back to the file. Following this, an additional function is executed to calculate the average metrics for each LLM, which also includes the original texts. A summary of these statistics is exported to a CSV file.

Before tokenizing the translated texts, preprocessing in the form of HTML parsing is necessary. The reason the translations are stored with their HTML tags is to preserve accurate formatting when used as cached data during webpage reloads. As shown in line 19 of Listing 6.1, the function `tokenize_german_text` not only returns a string of tokenized text but also produces a Python dictionary that counts the occurrence of clause dependents in the text. The Stanza Library is utilized to identify specific syntactic relations and clauses, employing dependency parsing and POS tagging for clause-dependent identification.

However, the clause dependent count results were excluded from the evaluation because many types of clause dependents recorded zero occurrences across the evaluated texts. This lack of relevant information made it impossible to conduct a significant evaluation for these metrics. Consequently, this data was not considered in the final analysis.

```
1  def evaluate_readability_for_file(filename):
2      """
3      Reads the original texts and the Easy Language translations
4      from a file, evaluates the readability and writes the
5      results back to the file.
6      The file has a JSON structure as follows:
7      {
8          website: "www.example.com",
9          model: "model_name",
10         translations: {
11             "original text": "easy language translation",
12             ...
13         }
14     }
15     """
16     # open file and read content from JSON
17     with open(filename, "r", encoding="utf-8") as file:
18         content = json.load(file)
```

```

18  # tokenize original text and translations
19  original_tokens, original_clause_dependents =
    tokenize_german_text(
20      content[Text.TRANSLATIONS.value].keys())
21
22  translations = [BeautifulSoup(
23      translation.replace("<br>", ""), "html.parser").get_text()
24      for translation in content[Text.TRANSLATIONS.value].values()
25  ]
26  translation_tokens, translations_clause_dependents =
    tokenize_german_text(translations)
27
28  # calculate readability results
29  readability_results_original =
    get_readability_scores(original_tokens)
30  readability_results_translations =
    get_readability_scores(translation_tokens)
31
32  # add results to json file
33  content["readability_results_original"] =
    readability_results_original
34  content["readability_results_original"][
35      Statistics.CLAUSE_DEPENDENTS.value
36  ] = original_clause_dependents
37  content["readability_results_translations"] =
    readability_results_translations
38  content["readability_results_translations"][
39      Statistics.CLAUSE_DEPENDENTS.value
40  ] = translations_clause_dependents
41  with open(filename, "w", encoding="utf-8") as file:
42      json.dump(content, file, indent=2, ensure_ascii=False)

```

Code Listing 6.1: Python Function to Tokenize and Evaluate Readability Metrics for a Given JSON File

The `get_readability_scores` function presented in Code Listing 6.2 calculates various readability metrics for a given text. It utilizes the `readability.getmeasures` function, which produces a dictionary containing numerous readability metrics and additional metrics, such as word count, sentence count and complex word count. From this output, only the LIX score is used directly. The other readability formulas relevant to this evaluation, such as the Amstad formula and Wiener Sachtextformel, are implemented separately within the script to ensure compatibility with the Ger-

man language.

```
1  def get_readability_scores(tokens):
2      """
3      Calculates the readability scores for a given text and returns
4      them as a dictionary.
5      Includes: Flesch Reading Ease German, 1. Wiener
6      Sachtextformel, G-SMOG Index, LIX, Sentence Count, Word
7      Count.
8      """
9      results = readability.getmeasures(tokens, lang=LANG)
10     FREG = flesch_reading_ease_german(results)
11     WSF = erste_wiener_sachtextformel(results, tokens)
12     GSMOG = smog_index_german(results)
13     readability_results = {
14         Indices.FLESCH_READING_EASE_GERMAN.value: FREG,
15         Indices.ERSTE_WIENER_SACHTEXTFORMEL.value: WSF,
16         Indices.G_SMOG_INDEX.value: GSMOG,
17         Indices.LIX.value: results["readability
18         grades"][Indices.LIX.value],
19         Statistics.SENTENCE_COUNT.value:
20             get_sentence_count(results),
21         Statistics.WORD_COUNT.value: get_word_count(results),
22     }
23     return readability_results
```

Code Listing 6.2: Python Function to Calculate Readability Metrics for a Tokenized Text

6.3 Results and Observations

The following subsections first analyze the readability results, based on the calculations from the evaluation script in Section 6.2, and then provide a spot check of the content quality of the generated translations, accompanied by relevant Figures.

6.3.1 Readability Results: Comparing the Three LLMs

Applying the four presented readability formulas to the Easy Language translations produced by the three LLMs, as well as to the original texts, revealed significant

improvements in readability and text structure. Table 6.2 provides a summary of the average scores across 20 evaluated websites listed in Appendix B, highlighting the improvements achieved by the LLMs, compared to the original content.

	Amstad	1. WSTF	gSmog	LIX	Sentence Count	Word Count
GPT-4o	69.5	3.6	1.5	39.1	123	1026
GPT-4o mini	69.5	3.8	2.0	38.6	157	1287
Llama 3.1 70B	67.1	4.2	2.8	40.1	142	1519
Original Texts	48.5	7.0	5.0	54.1	58	893

Table 6.2: Average Readability Results of Each LLM LLM translation to Easy Language and the Original Texts

All models showed significant gains on the Amstad index compared to the original texts, which had a score of 48.5. GPT-4o and GPT-4o Mini both achieved the highest scores at 69.5, while Llama 3.1 70B followed with a score of 67.1, showing a marginally lower improvement than the GPT models. The Amstad Index classifies texts with scores of 70 and above as Easy Language, whereas scores below 50 are considered difficult to read.

The results of the 1. WSTF indicate a notable reduction in linguistic complexity across all models. The original texts showed the highest complexity score of 7.0, aligning with a middle school reading level. GPT-4o achieved the greatest reduction, bringing the score down to 3.6, with GPT-4o Mini closely following at 3.8.

Llama 3.1 70B scored slightly higher at 4.2 but still demonstrated a significant improvement over the original content, lowering the difficulty to an elementary school level. It is important to note that, as defined in Section 6.1.2, the score typically starts at 4. One possible explanation for some results falling below this threshold could be the notably low average sentence length.

The gSmog index indicates a further reduction in linguistic complexity for the translations, assigning them significantly lower corresponding school grades compared to the original content. While the original texts were rated at a difficulty level of grade 5, all LLM models reduced this to below grade 3. GPT-4o achieved the most simplified structure with a score of 1.5, followed by GPT-4o Mini at 2.0 and Llama 3.1 70B at 2.8.

According to the LIX Index, all models simplified the original texts, which had an initial score of 54.1 and were classified as “difficult”. GPT-4o Mini achieved the lowest score at 38.6, falling into the “easy” category and matching grade 8 readability.

GPT-4o followed closely with a score of 39.1, also classified as “easy”. Llama 3.1 70B scored slightly higher at 40.1, placing it in the “medium” range, though this score lacks a specific classification under Bamberger’s interpretation. It is important to note that the LIX score is heavily influenced by the number of long words in the evaluated text. Most Easy Language texts still include longer words but often use a medial point to visually break them down. Without these longer words, the LIX scores would be even lower.

Structural changes are also evident in the Sentence Count and Word Count metrics. The original texts had an average of 58 sentences and 893 words. All models generated a longer output, with Llama 3.1 70B producing the highest word count (1519) and sentence count (142). GPT-4o Mini followed with 1287 words and 157 sentences, while GPT-4o created the most concise output among the models, with 1026 words and 123 sentences. Based on these metrics, an average sentence length can be derived, with the GPT models reducing length from 15 words to just 8, while the Llama model achieved an average of 10 words per sentence.

In summary, all three LLM models succeeded in simplifying the content and improving readability metrics, with GPT-4o excelling in most metrics. Because of that, the model was chosen for the second round of evaluation, this time comparing it against the already available Easy Language content from 20 other websites (see Appendix C).

6.3.2 Readability Results: Comparing GPT-4o with Provided Easy Language Texts

The results in Table 6.3 reflect a similar pattern to the results presented in Table 6.2 in terms of improving the original texts. However, this time, the original texts have better overall scores in all readability indices, resulting in slightly improved scores for the GPT-4o translations compared to those in Table 6.2. A notable difference between the GPT-4o translations and the provided Easy Language texts lies in sentence and word count. The generated translations once again increased these counts, averaging 72 sentences and 567 words, while the provided Easy Language texts maintained a sentence count similar to the original content but with fewer words. Comparing the average sentence length, GPT-4o was able to produce similar results with approximately 8 words per sentence — one word shorter than the average calculated for the Easy Language content. Both translations, however, successfully reduced the original average sentence length of 14 words.

	Amstad	1. WSTF	gSmog	LIX	Sentence Count	Word Count
GPT-4o	71.8	3.5	1.2	39.3	72	567
Easy Language	77.9	3.1	1.6	35.9	37	349
Original Texts	56.3	6.1	3.9	50.7	34	476

Table 6.3: Average Readability Results of GPT-4o, Easy Language Texts and the Corresponding Original Texts

6.3.3 Content Quality

The manual spot-check of translations revealed several content-related and technical issues stemming from the logic of content filtering and the translation process. One of the main challenges is the inconsistency in filtering out irrelevant content, such as headers and footers. Each website uses different naming conventions in its DOM structure, making it difficult to create a standardized method to identify and exclude these sections. In some instances, the add-on’s logic mistakenly overlooks paragraphs based on their HTML classes, while in other cases, it includes content that should have been excluded.

Another observed issue is inconsistent formatting within the translations. For example, while the prompt explicitly instructs the models to use the `<strong>` HTML tag for bold text, the output occasionally includes markdown-style formatting with `“**”`, which could cause confusion for the reader. Although the HTML formatting works correctly in many cases, this inconsistency affects the uniformity of the output. Furthermore, while important words are often marked in bold as intended, negations such as “not” are sometimes excluded from bold formatting, although the prompt specifies that negations must be emphasized.

An issue that appears primarily in translations produced by the Llama model is the occasional inclusion of special characters or letters from non-Latin alphabets that were not present in the original text. An example of this problem can be seen in Figure 6.2.

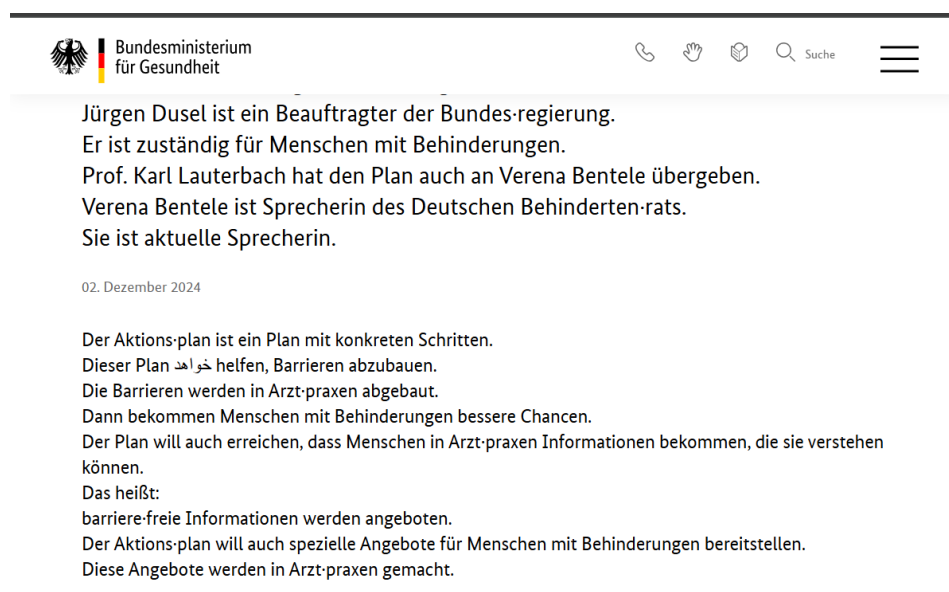


Figure 6.2: Arabic letters in translation by Llama 3.1 70B

As previously noted, the models produced translations with significantly higher sentence and word counts. Upon reviewing these translations, it is evident that the models tend to add explanatory sentences rather than condensing the content to its most essential points, as the prompt had instructed. While no false information was introduced, the models often repeated the same ideas or statements using different wording, without providing any new explanations or insights.

The technical challenges associated with handling diverse website structures are particularly evident when examining the three versions of an article from the German federal government website regarding the “Vertrauensfrage” (“vote of confidence”). A direct comparison of the original article³, the generated translation, and the Easy Language version provided by the website⁴ highlights the difficulty of identifying relevant content. For instance, the introduction in the original article (shown in Figure 6.3) was incorrectly filtered out by the add-on, resulting in not being translated by the LLM model (see Figure 6.4).

In contrast, the official Easy Language version in Figure 6.5 includes an Easy Language introduction. However, in the next paragraph, it deviates from the Easy Language guidelines by using a subordinate clause in the sentence “Der Bundeskanzler will wissen, ob ihm die Mitglieder vom Bundestag noch vertrauen.” The GPT-4o

³<https://www.bundesregierung.de/breg-de/suche/statement-antrag-vertrauensfrage-2324926>

⁴<https://www.bundesregierung.de/breg-de/leichte-sprache/leichte-sprache-abstimmung-vertrauens-frage-2327308>

translation handled this part more appropriately, rephrasing it as “Er will wissen: Wer ist für ihn? Wer ist gegen ihn?” Notably, while the GPT-4o translation reflects high linguistic accuracy overall, a grammatical error is evident in Figure 6.4 in the sentence “Am 11. Dezember hat Bundeskanzler Olaf Scholz die Vertrauensfrage beantragen”.

Despite this, the evaluated translations generally contained very few grammatical errors. When comparing the overall structure of the GPT-4o translation with the provided Easy Language article, an explanation for the difference in sentence and word count becomes apparent. The GPT-4o translation closely followed the original structure, while adding explanatory sentences or breaking up longer sentences. A possible explanation for this behaviour is discussed in Section 6.4. The Easy Language version, by contrast, offered a more concise version, focusing only on the most essential information.

Although the texts differ significantly in content, an analysis of the metrics calculated for this particular article in Table 6.4 reveals minimal differences between the generated translation, the Easy Language version, and the original text. Both translations contributed slight improvements to the original, which was already displaying metrics above average (see average metrics in Table 6.3). In some metrics, the GPT-4 generated translation showed a slight performance advantage. The increases in sentence and word counts align with the previously discussed phenomenon, where the language model tends to inflate these metrics. It is important to note that the introductory paragraph, which was excluded from translation, is not reflected in these counts.

	Amstad	1. WSTF	gSmog	LIX	Sentence Count	Word Count
GPT-4o	66.7	3.8	1.1	40.7	91	668
Easy Language	68.8	4.6	2.7	40.0	68	616
Original Text	62.7	4.8	2.5	42.9	74	658

Table 6.4: Readability Results for the Article Regarding the “Vertrauensfrage” (Original Article and Easy Language Article)



Im Wortlaut

Bundeskanzler Scholz beantragt Vertrauensfrage

Um den Weg für vorgezogene Bundestagswahlen frei zu machen, hat Bundeskanzler Olaf Scholz heute bei der Bundestagspräsidentin die Vertrauensfrage beantragt. Bis zur Bildung einer neuen Koalition blieben Bundesregierung und Bundestag voll arbeitsfähig, so der Kanzler.

🕒 3 Min. Lesedauer

Mitschrift Pressekonferenz

Mittwoch, 11. Dezember 2024



Am 11. Dezember hat Bundeskanzler Olaf Scholz die Vertrauensfrage beantragt.

Foto: Bundesregierung/Sandra Steins

Bundeskanzler Olaf Scholz:

Guten Tag, meine Damen und Herren! Soeben habe ich eine Abstimmung nach Artikel 68 des Grundgesetzes bei der Bundestagspräsidentin beantragt. Damit möchte ich den Weg für vorgezogene Bundestagswahlen frei machen. Meinen Antrag werde ich am kommenden Montag im Deutschen Bundestag ausführlich begründen. Wenn die Abgeordneten dann den von mir vorgeschlagenen Weg beschreiten, werde ich Bundespräsident Steinmeier am Montagnachmittag vorschlagen, den Bundestag aufzulösen. Folgt der Bundespräsident meinem Vorschlag, werden die Wählerinnen und Wähler am 23. Februar einen neuen Bundestag wählen können. Das ist mein Ziel.

In einer Demokratie sind es die Wählerinnen und Wähler, die den Kurs der künftigen Politik bestimmen. Sie entscheiden bei der Wahl, wie wir die großen Fragen beantworten, die vor uns liegen. Trauen wir uns zu, als starkes Land kraftvoll in

Figure 6.3: Original Article "Kanzler beantragt Vertrauensfrage"

1/7/25, 3:24 PM

Kanzler beantragt Vertrauensfrage | Bundesregierung



Im Wortlaut

Bundeskanzler Scholz beantragt Vertrauensfrage

Um den Weg für vorgezogene Bundestagswahlen frei zu machen, hat Bundeskanzler Olaf Scholz heute bei der Bundestagspräsidentin die Vertrauensfrage beantragt. Bis zur Bildung einer neuen Koalition bleiben Bundesregierung und Bundestag voll arbeitsfähig, so der Kanzler.

🕒 3 Min. Lesedauer

Mitschrift Pressekonferenz

Mittwoch, 11. Dezember 2024



Am 11. Dezember hat Bundeskanzler Olaf Scholz die Vertrauensfrage beantragen. Das ist ein wichtiger Schritt in der Politik.

Olaf Scholz will wissen:

Hat er das Vertrauen der Abgeordneten?

Das ist wichtig für seine Arbeit als Bundeskanzler.

Olaf Scholz will mit diesem Schritt Klarheit schaffen.

Er will wissen:

Wer ist für ihn?

Wer ist gegen ihn?

<https://www.bundesregierung.de/breg-de/suche/statement-antrag-vertrauensfrage-2324926>

1/4

Figure 6.4: Easy Language Translation by GPT-4o of The Article "Kanzler beantragt Vertrauensfrage"

07.01.25, 15:29

Leichte Sprache: Vertrauens-Frage: Weg frei für Neu-Wahlen | Bundesregierung



Bundes-Kanzler Olaf Scholz hat die Vertrauens-Frage gestellt

Jetzt kann es neue Wahlen geben

Die Menschen in Deutschland sollen einen neuen Bundestag wählen.

Deshalb hat Bundes-Kanzler Olaf Scholz

im Bundestag die Vertrauens-Frage gestellt.

Das Ergebnis:

Die Mehrheit von den Abgeordneten vertraut

dem Bundes-Kanzler **nicht** mehr.

Der Bundes-Präsident hat den Bundestag aufgelöst.

Freitag, 27. Dezember 2024



Bundes-Kanzler Olaf Scholz hält eine Rede zur Vertrauens-Frage im Bundestag.

Foto: Bundesregierung/Steffen Kugler

Der Bundes-Kanzler Olaf Scholz hat die Vertrauens-Frage
im Deutschen Bundestag gestellt.

Vertrauens-Frage bedeutet:

Der Bundes-Kanzler will wissen,

ob ihm die Mitglieder vom Bundestag noch vertrauen.

Und ob er Bundes-Kanzler bleiben soll.

Die Mitglieder vom Bundestag stimmen dann darüber ab.

Die Mitglieder vom Bundestag nennt man Abgeordnete.

Figure 6.5: Easy Language Article "Jetzt kann es neue Wahlen geben"

6.4 Discussion

The evaluation results show that while readability improved across all translations, the content often became significantly longer. This increase in length was mainly due to one-to-one translations of the original content, with only minor explanations added, instead of condensing the information to its most relevant points.

A key observation, especially on ministry websites, is that the Easy Language content provided by these sources differs substantially from the regular versions of the same pages. The Easy Language webpages are often designed with accessibility in mind. This includes simplified structures, rearranged text, carefully selected images, and a prioritization of key information. In contrast, the standard versions focus on different content and layout priorities, which are generally less accessible. Additionally, the original texts are often written with the assumption that readers possess some prior knowledge of the topic. For the browser add-on, this presents a challenge: it translates directly from the regular version of the webpage, which serves as a different foundation than the curated Easy Language versions provided by some websites. As a result, the generated translations lack focus on key information and a simplified structure compared the provided Easy Language pages. Some formatting errors, such as inconsistent bold text or incorrect special characters, can be corrected through improvements in the code.

However, content-related issues, such as ensuring concise translations that focus on the most essential points, are much harder to manage. One potential improvement to the translation process would be to feed the entire relevant content of a webpage into a single translation prompt, rather than translating each paragraph individually. Providing the model with the full context at once would enable it to make more informed decisions about which information is most essential and which can be discarded. However, this approach may be limited by the constraints of the model's context window, which includes both the input size and the generated output. If the website's content is too long, the combined size of the input and the translated output could exceed the model's context window, leading to incomplete information.

To test this hypothesis, the spot-checked article in Figure 6.3 was translated again using GPT-4o. This time, the translation was performed in the OpenAI's interactive chat⁵, where the entire content of the webpage was submitted at once. The test was conducted twice in separate chat sessions. While the number of words and sentences decreased to 489 and 63 in the first chat and 477 and 56 in the second, both even lower than the provided Easy Language version, the overall content remained largely

⁵<https://chatgpt.com>

similar to the translations produced with the paragraph-by-paragraph approach by the browser add-on.

Ultimately, while the add-on can assist with translation, it cannot fully replace human translations, especially given that it lacks the general understanding of the source text that a translator would gain when directly communicating with the author.

Chapter 7

Conclusion

The goal of this thesis was to develop a browser add-on which is able to automatically translate any web text into German Easy Language. By using external APIs and Large Language Models (LLMs) such as GPT and Llama to generate translations, the add-on proved to improve the readability metrics of the evaluated web texts to an Easy Language level and demonstrates its significant potential as an accessibility tool. The readability scores of the translations generated by LLMs are similar to the official Easy Language texts provided by the evaluated websites, which indicates that the generated texts are just as accessible as the manually translated versions. By allowing users to access Easy Language content on demand, such as government websites or news articles, the add-on fills a critical gap in online content and offers a practical solution for people with cognitive impairments, who rely on simplified language to navigate the Web.

Despite these strengths, the add-on deals with certain limitations that must be taken into account. Since the translations are not reviewed by human experts, there is an increased risk of inaccuracies or misinterpretations. This limitation underscores a broader challenge of translations generated by Artificial Intelligence (AI): they cannot match the contextual understanding and reliability that human translators offer. Additionally, original texts are often written with the assumption that readers possess some prior knowledge of the topic, which increases the likelihood of misunderstandings in the generated Easy Language translations. Nevertheless, the add-on has proven effective in improving readability metrics. Translations generated by the tool typically offer a clearer structure and more accessible language.

From a technical perspective, filtering content for translation represents a significant challenge. Each website has unique conventions for structuring content, with dif-

ferences in HTML tags, class names, and content placement. The add-on addresses this by implementing a filtering mechanism to exclude irrelevant elements, such as navigation menus or references. While this approach occasionally leads to false negatives, it significantly enhances the quality of the translated content and optimizes token usage. This makes the tool both cost-effective and efficient. The reliance on external APIs introduces critical factors such as token limits, service reliability, and potential pricing model changes, all of which can affect the add-on's performance. However, using external APIs has proven to be more efficient than hosting an LLM locally, particularly in terms of integration and requirements on the users devices.

Ultimately, the add-on is a supportive tool rather than a replacement for manually created Easy Language texts. Human expertise remains crucial for ensuring accuracy, contextual relevance, and overall quality. However, with LLM technology constantly advancing, the add-on's capabilities are expected to improve.

In conclusion, the add-on fills the gap of accessible online content by providing Easy Language translations for any given website. By utilizing LLMs like GPT and Llama, it demonstrates the potential of AI to enhance inclusivity, while also underscoring the importance of careful usage and human oversight. Despite its limitations, this work represents a valuable step forward in making digital information more inclusive.

7.1 Outlook

The browser add-on has laid a solid foundation, but there are several areas for improvement to enhance both its functionality and user experience.

As the discussion highlights, the prompt can still be optimized. Various approaches should be tested to improve the LLM's ability to produce concise translations with a structure that differs from the original text. Considering this challenge from a different perspective, fine-tuning an open-source LLM on Easy Language data is another opportunity to improve the translation quality and eliminate the need to list a complete guideline in the prompt. Another important step is to manage the fine-tuned LLM in a dedicated backend to be independent from external APIs. All issues concerning rate limits and cost would dissolve and it would enhance efficiency.

Expanding browser compatibility is another crucial improvement, as the add-on currently only supports Chrome. Adapting it to other browsers, such as Firefox and Edge, would make it more widely accessible. Enhancing the visual presentation of translations is also an important consideration. The feature of automatically

adjusting font sizes for translated text was not implemented in the browser add-on due to inconsistent behavior across different websites. However, this improvement could be considered in the future, as it would significantly enhance readability, particularly for users with visual impairments or cognitive disabilities. To support the message of the translated texts, difficult concepts could be illustrated with automatically selected or generated images, which is an Easy Language rule that is currently not included in the prompt or the add-on's implementation. With these visual and functional changes planned, testing the add-on with the target group would offer valuable insights into its usability and accessibility.

By addressing these areas for improvement, the add-on can overcome its current limitations and become an even more useful tool for making digital information accessible to all users.

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Appendix A

German Easy Language Rules

1. Zur Verfügung stehen folgende Sonderzeichen: Punkt, Frage-, Ausrufezeichen, Doppelpunkt, Anführungszeichen, Mediopunkt/Mittelpunkt (·). Das Komma gehört damit nicht zum System der Leichten Sprache
2. Der Doppelpunkt leitet Aufzählungen ein
3. Zahlen werden als Ziffern geschrieben.
Beispiel: 20 Hunde anstatt zwanzig Hunde. Achtung: „ein“ als indefiniter Artikel ist kein Zahlwort.
4. Der Mediopunkt kennzeichnet innerhalb eines zusammengesetzten Worts die Grenze zwischen zwei Einzelwörtern und macht sie dadurch besser erkennbar.
Beispiel: Lotto-Annahme-stelle
5. Jahreszahlen, Prozentzahlen und hohe Zahlen dürfen stehen bleiben wenn sie für den Text zentral sind (Geschichtstext, Steuererklärung). Ansonsten Redewendungen wie: vor langer Zeit, vor über 100 Jahren, sehr viel usw.
6. Grundwortschatz verwenden **Beispiel:** Vogel anstatt Gartengrasmücke
7. möglichst kurze Wörter verwenden. Wenn nicht möglich, dann lange Wörter lesbarer machen mithilfe des Mediopunkts (**Beispiel:** Kranken·haus)
8. Fach- und Fremdwörter vermeiden oder (sofern für den Text zentral) erklären
9. Schriftbasierte Abkürzungen („usw.“, „s. u.“) vermeiden. Bekannte Siglenwörter („LKW“) dürfen eingesetzt werden.

10. Verbal statt nominal: Nominalstil vermeiden; Auflösung mit „Dreischritt der Übersetzung“
Beispiel: Eine Messung durchführen > messen
11. Passiv vermeiden. Handlungsträger ermitteln und hinzufügen
Beispiel: Heute wählen wir den Heim·beirat.
12. Genitive auflösen mit "von"-Periphrase. Prüfe ob der Genitiv in eine verbale Struktur überführt werden kann
Schlecht: Des Lehrers Haus. Das Haus des Lehrers.
Gut: Das Haus vom Lehrer
Schlecht: Deutschlands Aufarbeitung der eigenen Geschichte...
Gut: Deutschland muss die eigene Geschichte aufarbeiten. Dann . . .
13. Satzgliedstellung Subjekt-Prädikat-Objekt (S-P-O), wenn möglich.
14. Nur eine Aussage pro Satz
15. Keine Nebensätze. Auflösung von Satzgefügen gemäß den Vorgaben.
16. Konditionalsatz: Wenn...dann... wird zu einer Frage und einem Satz beginnend mit "Dann". Auch Sätze mit "Sofern" "falls, (so)" usw. werden aufgelöst.
Schlecht: Wollen Sie einen Nachteilsausgleich in Anspruch nehmen, (so) müssen Sie einen Antrag stellen.
Gut: Sie möchten einen Nachteils·ausgleich haben? Dann müssen Sie einen Antrag stellen.
17. Kausalsatz: „weil“, „da“, „zumal“ werden zu "deshalb" oder "nämlich"
Schlecht: Die Eisdiele musste im Winter schließen, weil keine Kunden mehr kamen.
Gut: Es waren keine Kunden mehr da. Deshalb musste die Eisdiele im Winter schließen.
Gut: Die Eisdiele musste im Winter schließen. Es waren nämlich keine Kunden mehr da.
18. Modalsatz: : „dadurch dass“, „indem“, „sodass“ wird zu "so"
Schlecht: Lina hat Muskelmasse aufgebaut, indem sie jeden Tag Sport getrieben hat
Gut: So ist Lina stark geworden: Lina hat jeden Tag Sport gemacht.
19. Temporalsatz: : „während“, „bis“, „als“, „seit“, „nachdem“, „bevor“, „solange“ wird zu Sätzen in chronologischer Reihenfolge mit "Dann" "jetzt" oder "nun".

Schlecht: Sie wurde schwanger und dann haben sie geheiratet.

Gut: Maria wurde schwanger. Deshalb haben Maria und Peter dann bald geheiratet.

Gut: Ralf hat 3 Bier getrunken. Dann ist Ralf trotzdem noch Auto gefahren.

20. Konsekutivatz: „sodass“, „so . . . , dass“ wird zu "deshalb"

Schlecht: Es kamen nur noch so wenige Kunden, dass die Eisdiele schließen musste.

Gut: Es sind nur noch sehr wenige Kunden gekommen. Deshalb musste die Eisdiele schließen.

21. Konzessivsatz: „obwohl“, „obgleich“, „obschon“, „wenngleich“, „wenn auch“ wird zu "trotzdem"

Schlecht: Obwohl Lars sich sehr gründlich auf die Prüfung vorbereitet hat, hat er sie nicht bestanden.

Gut: Lars hat sich sehr gut auf die Prüfung vorbereitet. Lars hat die Prüfung trotzdem nicht bestanden.

22. Finalsatz: „damit“, „dass“, „auf dass“, „um zu“ wird zu "wollen" + "deshalb"

Schlecht: Er schoss den Ball in die linke Ecke, um den Torwart zu verwirren.

Gut: Er wollte den Torwart verwirren. Deshalb hat er den Ball in die linke Ecke vom Tor geschossen.

23. Relativsatz: Aufteilung in mehrere Aussagen oder notwendige Informationen erläutern.

Schlecht: Zeugen, die bei der Vernehmung schweigen, sind unbrauchbar.

Gut: Die Zeugen sagen bei der Vernehmung nichts? Dann sind es schlechte Zeugen.

Schlecht: Die Zeugen, die übrigens bei der Vernehmung schwiegen, stellten sich als unbrauchbar heraus.

Gut: Die Zeugen waren nicht gut. Die Zeugen haben nämlich bei der Vernehmung nicht gesprochen.

24. Komplexe nominale Strukturen zunächst in ein Satzgefüge umformen und dann in einzelne Aussagen auflösen

Schlecht: Wir haben Ihnen die Handhabung Ihrer Hörgeräte bereits umfassend erläutert.

Gut: Als Satzgefüge: Wir haben Ihnen bereits umfassend erläutert, wie Ihre Hörgeräte funktionieren / wie Sie Ihre Hörgeräte benutzen müssen. In leichter Sprache: Die Mitarbeiter von [Firma A] Hörgeräte haben Ihnen die

Hörgeräte erklärt: So müssen Sie die Hörgeräte benutzen. So funktionieren die Hörgeräte.

25. Konjunktiv sofern möglich vermeiden. Aussagen reformulieren. Indirekte Rede zur direkten Rede formulieren.

Schlecht: Wenn Sie einen Antrag gestellt hätten, dann wäre Ihnen ein Nachteilsausgleich zugesprochen worden.

Gut: Für einen Nachteilsausgleich müssen Sie einen Antrag stellen. Der Antrag musste bis zum ... beim Amt sein. Sie haben keinen Antrag gestellt. Deshalb bekommen Sie keinen Nachteilsausgleich.

Schlecht: Es wäre schön, wenn er käme.

Gut: Er kommt vielleicht? Das würde uns freuen.

26. Präteritum vermeiden. Stattdessen Perfekt oder Präsens mit Rahmensetzung. Ausnahme: Hilfsverben und Modalverben (wollte, waren, durften, usw.)

Schlecht: Ich aß

Gut: Ich habe gegessen

Gut: Es ist das Jahr 1525. Die Bauern führen Krieg gegen die Fürsten. Noch steht Martin Luther auf der Seite der Bauern.

27. Futur tendenziell vermeiden. Präsens setzen.

Schlecht: Er ist ein hervorragender Musiker und er wird sich mit großem Engagement seinen Aufgaben in Frankfurt widmen (-> Er arbeitet noch nicht in Frankfurt).

Gut: Edward ist ein sehr guter Musiker. Edward arbeitet bald in Frankfurt. Edward macht dann seine Aufgaben sehr gut.

Schlecht: Jetzt wird niemand zu Hause sein.

Gut: Jetzt ist sicher niemand zu Hause.

28. Transparente Metaphern dürfen verwendet werden. Weniger bekannte ersetzen oder erläutern.

Schlecht: Er ist ein Löwe.

Gut: Peter ist so stark wie ein Löwe.

Gut: Peter sagt: „Ralf und Maria sind Raben-eltern.“ Raben-eltern heißt: Ralf und Maria sind schlechte Eltern.

29. Negation nach Möglichkeit vermeiden. Nach Möglichkeit mit "nicht" negieren und fett im Text setzen.

Schlecht: Es darf keine Neigung zu Sehnenscheidenentzündungen vorhanden sein

Gut: Sie müssen gesunde Hände haben.

Schlecht: Wir haben heute keinen Kuchen gebacken.

Gut: Wir haben heute nicht Kuchen gebacken. Oder: Das haben wir heute nicht gemacht: Kuchen backen.

30. Textuelle Entfaltung beachten. – Was soll ausgesagt werden? – Welches sind die zentralen Gedanken? – Welches sind die Abschnitte und Teile des Texts? – Wie verhalten sich die einzelnen Aussagen zueinander? – Wo verläuft der argumentative Faden und was ist Erklärung, Erläuterung, Beispiel?

31. Für alle Wortarten gilt: Verwendung gleicher Wörter für gleiche Sachverhalte, keine Synonyme

Beispiel: Sie schreiben über ein Medikament. Benutzen Sie immer ein Wort. Zum Beispiel: Tablette. Wechseln Sie nicht zwischen Tablette und Pille.

32. Personalpronomen der 1. und 2. Person können verwendet werden.

Schlecht: Lisa sagt, sie sei krank.

Gut: Lisa sagt: "Ich bin krank".

Schlecht: Heute wird der Vorstand gewählt.

Gut: Heute wählen wir den Vorstand.

Schlecht: Für die Gewährung der Mittel muss ein Antrag gestellt werden.

Gut: Sie möchten die Mittel bekommen? Dann müssen Sie einen Antrag stellen.

33. Personalpronomen der 3. Person müssen aufgelöst werden. Sie werden durch das Nomen ersetzt, für das sie stehen.

Schlecht: Der Vater kaufte dem Mädchen ein Eis, um ihm eine Freude zu machen.

Gut: Der Vater hat dem Mädchen ein Eis gekauft. Der Vater wollte dem Mädchen damit eine Freude machen.

34. Das expletive Es, das kein Bezugswort aufweist, darf verwendet werden.

Beispiel: Es regnet.

35. Erwachsene Leserinnen und Leser werden in der Regel gesiezt.

36. Bei Verben des Sagens und Denkens: Wechsel in direkte Rede.

Schlecht: Peter sagte am Telefon, er sei krank.

Gut: Peter sagte am Telefon: Ich bin krank.

Schlecht: Manche Menschen haben so viel Geld, dass sie nicht wissen, wie sie es ausgeben sollen.

Gut: Manche Menschen haben sehr viel Geld. Viele von diesen Menschen wissen nicht: Wie sollen wir das Geld ausgeben?

37. Zwischenüberschriften setzen.

38. Nach einem schweren Wort erfolgt die Erklärung, erkennbar durch eine Einrückung in der nächsten Zeile

Beispiel:

Herr Meier hatte einen schweren Unfall.

Jetzt macht Herr Meier eine berufliche Rehabilitation.

Das heißt:

Herr Meier lernt einen anderen Beruf

39. Bei Übersetzungen in Leichte Sprache darf der Text verändert werden (Abschnitte, Überschriften, etc.). Innerhalb eines Absatzes kann die Information umgestellt werden.

40. Haftungsausschluss für Texte, deren Ausgangsversion rechtsverbindlich war.

Beispiel:

Der Text in Leichter Sprache soll Sie nur informieren.

Der Text ist nur ein Zusatzangebot.

Der rechts-gültige Text ist das Gesetz.

Der Text in Leichter Sprache ist rechts-unwirksam.

Das bedeutet:

Mit dem Text in Leichter Sprache können Sie nicht einen Nachteils-ausgleich einfordern.

41. Leichte-Sprache-Texte sind Listen: Jeder Satz beginnt mit einer neuen Zeile
Einrückungen für Erläuterungen und Beispiele verwenden

Beispiel:

Sie müssen zu dem Gerichts-termin kommen.

Sie haben vielleicht schon eine Aussage gemacht.

Zum Beispiel:

- Bei der Polizei.
- Oder vor Gericht.
- Oder bei einer Staats-anwaltschaft.
- Oder in einer früheren Haupt-verhandlung.

Sie müssen trotzdem zu diesem Gerichts-termin kommen.

Das ist sehr wichtig.

42. Tendenziell serifenlose Schriftarten verwenden. -> hardcoded in add-on
43. Hervorhebung nur durch **Fettdruck**.
44. Abtrennung im Satz nach syntaktischen Gruppen.
Schlecht: Der letzte Urlaub auf Mallorca war ein Erlebnis.
Gut: Der letzte Urlaub auf Mallorca
war ein Erlebnis.
45. Einsatz von Bildern:
zentrale Konzepte bebildern, Aussparen der Illustrationsfunktion
Einsatz von Fotos, Piktogrammen, Diagrammen
keine kindlich anmutenden Bilder für erwachsene Adressatinnen und Adressaten
Potential der Multicodalität ausschöpfen
Achtung Bildrechte!

Appendix B

Evaluated Websites

All websites were last accessed on 7 January 2025.

- <https://soziales.hessen.de/presse/landesregierung-schnue-rt-massnahmenpaket-zur-staerkung-fruehkindlicher-bildung>
- <https://soziales.hessen.de/Betreutes-Wohnen>
- <https://soziales.hessen.de/frauen/hilfe-bei-gewalt/und-was-tust-du>
- <https://www.bundesregierung.de/breg-de/suche/statement-antrag-vertrauensfrage-2324926>
- https://www.bmj.de/SharedDocs/Pressemitteilungen/DE/2024/1211_Mietpreisbremse.html
- https://www.bmj.de/DE/rechtsstaat_kompakt/grundgesetz/einleitung_grundgesetz/einleitung_grundgesetz_node.html
- <https://www.bundesgesundheitsministerium.de/presse/pressmitteilungen/aktionsplan-barrierefreies-gesundheitswesen-pm-02-12-24.html>
- <https://www.bundesgesundheitsministerium.de/gkv.html>
- <https://www.bafza.de/engagement-und-aktionen/aktion-zusammen-wachsen>
- <https://www.bafza.de/engagement-und-aktionen/menschen-staerken-menschen>

- <https://www.tagesschau.de/inland/bundestagswahl/wahlprogramme-bundestagswahl-2025-100.html>
- <https://www.tagesschau.de/inland/deutschlandticket-einigung-100.html>
- <https://www.tagesschau.de/inland/gesellschaft/dwd-wetter-jahresbilanz-100.html>
- <https://www.tagesschau.de/wissen/gesundheit/grippe-schutz-impfung-100.html>
- <https://www.hessenschau.de/panorama/falsche-qr-codes-polizei-warnt-vor-neuer-betrugsmasche-quishing-v1,betrugsmasche-quishing-100.html>
- <https://www.hessenschau.de/freizeit/kostenlose-ausflugsziele-in-hessen-die-trotzdem-begeistern-v2,kostenlos-und-draussen-100.html>
- <https://www.tagesschau.de/ausland/europa/deutsche-delegation-damaskus-100.html>
- <https://www.tagesschau.de/inland/regional/bayern/br-klub-der-gewinner-zeigt-gesicht-gegen-mobbing-104.html>
- <https://www.tagesschau.de/inland/innenpolitik/kanzlerkandidaten-bundestagswahl-100.html>
- <https://www.tagesschau.de/inland/gesellschaft/deutsche-bahn-aenderungen-winterfahrplan-100.html>

Appendix C

Evaluated Websites and Provided Easy Language Versions

All websites were last accessed on 9 January 2025.

- <https://www.bundesregierung.de/breg-de/suche/novelle-bau-gesetzbuch-2306066>
<https://www.bundesregierung.de/breg-de/leichte-sprache/tempo-beim-wohnungsbau-2306532>
- <https://www.ndr.de/nachrichten/info/Erneut-Sturmflutwarnung-fuer-Nordseekueste-und-Hamburg,wetter9950.html>
https://www.ndr.de/fernsehen/barrierefreie_angebote/leichte_sprache/Starker-Sturm-in-Norddeutschland,sturm4276.html
- <https://www.ndr.de/nachrichten/info/Bahn-und-GDL-verhandeln-wieder-vorerst-keine-Streiks-mehr,bahn3296.html>
https://www.ndr.de/fernsehen/barrierefreie_angebote/leichte_sprache/Einigung-bei-der-Bahn,lokfuehrer226.html
- <https://www.mdr.de/nachrichten/sachsen/leipzig/fake-plakat-bau-eisenbahnstrasse-parkplaetze-lene-voigt-park-100.html>
<https://www.mdr.de/nachrichten-leicht/leichte-sprache-sachsen-leipzig-fake-plakate-lene-voigt-park-100.html>

- <https://www.mdr.de/nachrichten/sachsen/dresden/freital-pirna/elbe-bruecke-gesperret-behelfsbruecke-vermessung-bad-schandau-100.html>
<https://www.mdr.de/nachrichten-leicht/leichte-sprache-sachsen-bad-schandau-vermessung-behelfsbruecke-100.html>
- <https://www.mdr.de/nachrichten/sachsen/bilanz-silvester-nacht-tote-verletzte-braende-100.html>
<https://www.mdr.de/nachrichten-leicht/leichte-sprache-silvester-mitteldeutschland-100.html>
- <https://www.mdr.de/nachrichten/sachsen-anhalt/magdeburg/magdeburg/polizei-einsatz-veterinaeramt-tote-tiere-102.html>
<https://www.mdr.de/nachrichten-leicht/leichte-sprache-sachsen-anhalt-magdeburg-tote-tiere-100.html>
- <https://www.mdr.de/nachrichten/thueringen/mario-voigt-neujahrsansprache-ministerpraesident-100.html>
<https://www.mdr.de/nachrichten-leicht/leichte-sprache-neujahrs-ansprache-thueringen-100.html>
- <https://www.mdr.de/nachrichten/sachsen-anhalt/post-streik-tarifverhandlungen-102.html>
<https://www.mdr.de/nachrichten-leicht/leichte-sprache-sachsen-anhalt-post-streikt-100.html>
- <https://www.mdr.de/nachrichten/thueringen/ost-thueringen/altenburg/unterkunft-fluechtlinge-rassistisch-attacke-schmoelln-100.html>
<https://www.mdr.de/nachrichten-leicht/leichte-sprache-thueringen-schmoelln-attacke-unterkunft-100.html>
- <https://www.mdr.de/nachrichten/thueringen/sternsinger-kinder-eichsfeld-nordhausen-100.html>
<https://www.mdr.de/nachrichten-leicht/leichte-sprache-thueringen-sternsinger-weniger-kinder-100.html>
- <https://www.bundesregierung.de/breg-de/suche/telefonische-krankschreibung-1800026>
<https://www.bundesregierung.de/breg-de/leichte-sprache/krank-schreibung-2248356>

- <https://www.bundesregierung.de/breg-de/suche/menschen-mit-behinderung-2246674>
<https://www.bundesregierung.de/breg-de/leichte-sprache/wichtige-schritte-fuer-die-inklusion-2324272>
- <https://www.bundesregierung.de/breg-de/suche/statement-antrag-vertrauensfrage-2324926>
<https://www.bundesregierung.de/breg-de/leichte-sprache/leichte-sprache-abstimmung-vertrauens-frage-2327308>
- <https://www.bmi.bund.de/SharedDocs/pressemitteilungen/DE/2023/06/global-forum.html>
<https://www.bmi.bund.de/SharedDocs/pressemitteilungen/DE/2023/06/global-forum-LS.html>
- https://www.aschaffenburg.de/Aktuelles/Aktuelle-Meldungen/DE_index_4160_98540.html
https://www.aschaffenburg.de/Aktuelles/Aktuelle-Meldungen-in-Leichter-Sprache/DE_index_6736_98544.html
- https://www.aschaffenburg.de/Aktuelles/Aktuelle-Meldungen/DE_index_4160_98518.html
https://www.aschaffenburg.de/Aktuelles/Aktuelle-Meldungen-in-Leichter-Sprache/DE_index_6736_98520.html
- https://www.aschaffenburg.de/Aktuelles/Aktuelle-Meldungen/DE_index_4160_98491.html
https://www.aschaffenburg.de/Aktuelles/Aktuelle-Meldungen-in-Leichter-Sprache/DE_index_6736_98498.html
- https://www.aschaffenburg.de/Aktuelles/Aktuelle-Meldungen/DE_index_4160_98161.html
https://www.aschaffenburg.de/Aktuelles/Aktuelle-Meldungen-in-Leichter-Sprache/DE_index_6736_98195.html
- https://www.aschaffenburg.de/Buerger-in-Aschaffenburg/Strasse-und-Verkehr/Radverkehr/Stadtradeln/DE_index_4748.html
https://www.aschaffenburg.de/Aktuelles/Aktuelle-Meldungen-in-Leichter-Sprache/DE_index_6736_98134.html